Spanish–Asturian Neural Machine Translation

WMT24 Low-Resource MT Shared Task — Extended Abstract

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Abstract

We describe three systems built for the WMT24 Spanish–Asturian MT task. Starting from a MarianMT bilingual baseline (16.52 BLEU), we move to `nllb-200-distilled-600M` with LoRA adaptation on the full ES-AST corpus, reaching **22.28 BLEU** — above the Apertium rule-based baseline (17.0) and comparable to the best constrained WMT24 submissions (≈22). A continual multilingual fine-tuning experiment with Galician and Portuguese triggered catastrophic forgetting (14.72 BLEU), which we analyse as a substantive finding. All experiments were run locally on an NVIDIA RTX 4060 using Python scripts.

1. Approach and Justification

The constrained track [1] restricts pre-trained models to $\leq$1B parameters. We follow the fine-tuning approach, selecting `facebook/nllb-200-distilled-600M` [2] (600M params) as the base model. NLLB was chosen because it natively supports Asturian (`ast_Latn`) and provides multilingual representations across 200 languages, giving fine-tuning a stronger initialisation than a bilingual checkpoint.

The key efficiency decision was **LoRA** [3] (r=16, α=32, applied to all attention projections), which reduces trainable parameters to $\sim$0.34% of the base model and makes full-corpus training feasible on an 8GB GPU. NLLB's built-in target language token (`ast_Latn`) serves as the forced BOS at generation time, handling multilingual routing without vocabulary modification.

2. Data

ES-AST. Projecte AINA Spanish–Asturian parallel corpus [5]: $\sim$704k pairs. System 1 used 100k; System 2 used the full corpus (697,334 train, val capped to 4,000).

ES-GL. 50k pairs from SciELO-GL [6] (`es-gl`) — a biomedical parallel corpus. System 3 only.

ES-PT. 1,327 pairs from `opus_books` [7] (`es-pt`) — the full available split. System 3 only.

Note on constraints. Systems 1 and 2 use only the primary shared task corpus and are fully within the constrained track. System 3 additionally uses `SciELO-GL` and `opus_books`, which are not listed as official shared task resources; it should be considered an additional out-of-constraint experiment.

Evaluation. All systems evaluated on FLORES+ `devtest` [8]: 1,012 held-out `spa_Latn`$\rightarrow$`ast_Latn` pairs. SacreBLEU [9] throughout.

3. Systems

3.1. System 1 — MarianMT Bilingual Baseline

`Helsinki-NLP/opus-mt-es-ca` [4] fine-tuned on 100k ES-AST pairs for one epoch. Catalan was chosen as the source checkpoint because it shares a SentencePiece vocabulary and Western Romance structure with Asturian. Training: Adafactor (lr 1×10^{-4}), bf16, gradient checkpointing.

3.2. System 2 — NLLB + LoRA (best system)

LoRA fine-tuning of NLLB-200-distilled-600M on the full ES-AST corpus (697,334 train / 4,000 val) for one epoch. Training: Adafactor (lr 5×10^{-4}), bf16, gradient checkpointing, effective batch 16. Runtime: $\sim$9 hours on RTX 4060.

3.3. System 3 — Continual Multilingual Fine-Tuning

The System 2 adapter was continued on 50k GL + 1,327 PT + 10k AST for one epoch (lr 2×10^{-4}; total 61,327 pairs). Target-language tokens (`glg_Latn`, `por_Latn`, `ast_Latn`) condition training labels and serve as the forced BOS token during generation.

4. Results

Table 1: SacreBLEU on FLORES+ `devtest` (1,012 sentences).

System	BLEU
Zero-shot `opus-mt-es-ca`	3.24
Apertium rule-based [1]	17.0
Sys. 1: MarianMT, 100k AST	16.52
Sys. 2: NLLB + LoRA, 704k AST	**22.28**
Sys. 3: continual ML (GL + PT + AST)	14.72
Best WMT24 constrained [1]	≈22

System 2 beats the Apertium rule-based baseline by +5.28 BLEU and places in the same range as the best constrained WMT24 submissions. System 1 falls just below Apertium, reflecting the limited signal from 100k pairs and one epoch. System 3 regresses sharply — discussed below.

5. Discussion

System 2. The bulk of the gain over System 1 comes from switching to NLLB, whose multilingual pre-training already encodes Asturian-adjacent representations. LoRA made the full 704k corpus tractable on an 8GB GPU while updating only ∼0.34% of parameters. System 1's data limitation explains why it barely clears the zero-shot output and fails to match even the rule-based baseline.

Catastrophic forgetting (System 3). The -7.56 BLEU drop is a clear case of catastrophic forgetting [10]. The training ratio was ∼51k GL+PT versus 10k AST (≈5:1), insufficient to anchor Asturian representations. A secondary factor is domain mismatch: SciELO-GL is biomedical and `opus_books` is literary, both distant from FLORES+'s general-domain sentences. System 3 is nonetheless the most genuinely multilingual design of the three; the forgetting result is a substantive finding that directly replicates patterns well-documented in the continual learning literature.

6. Limitations and Future Work

The immediate fix for System 3 is **balanced sampling** (equal AST/GL/PT proportions) rather than the 5:1 imbalance used here. **Adapter merging** [11] offers an alternative: combining adapters post-hoc avoids joint training and forgetting entirely. Further gains on System 2 are likely from additional epochs, a larger LoRA rank (r=32), and replacing SciELO-GL with a general-domain Galician corpus better matched to FLORES+ evaluation conditions.

Submitted Files

All files are submitted as a single zip archive (`kerins_spa_ast_mt.zip`).

- `finetune_es_ast.py` — System 1 (MarianMT)
- `finetune_multilingual_ast.py` — System 2 (NLLB + LoRA)
- `continue_multilingual.py` — System 3 (continual ML)
- `export_flores_peft.py` — FLORES+ hypothesis generation
- `flores_devtest_hypotheses_bleu22.28.txt` — System 2 outputs (best)
- `flores_devtest_hypotheses_bleu14.72.txt` — System 3 outputs
- `train_run.log` — System 2 log (verifies BLEU 22.28)

References

[1] WMT24 Organizers. 2024. *Findings of the WMT24 Shared Task on Low-Resource MT*. *Proc. WMT*. https://aclanthology.org/2024.wmt-1.57/

[2] NLLB Team. 2022. *No Language Left Behind*. arXiv:2207.04672.

[3] E. Hu et al. 2022. *LoRA: Low-Rank Adaptation of Large Language Models*. *Proc. ICLR*.

[4] J. Tiedemann and S. Thottingal. 2020. *OPUS-MT*. *Proc. EAMT*.

[5] Projecte AINA. 2023. *ES-AST Parallel Corpus*. Hugging Face. https://hf.co/datasets/projecte-aina/ES-AST_Parallel_Corpus

[6] Z. Jaume et al. *SciELO-GL* [Dataset]. https://hf.co/datasets/ZJaume/SciELO-GL

[7] J. Tiedemann. 2012. *Parallel Data, Tools and Interfaces in OPUS*. *Proc. LREC*. `opus_books` via Hugging Face.

[8] NLLB Team / Open Language Data Initiative. 2022. *FLORES+* [Dataset]. https://hf.co/datasets/openlanguagedata/flores_plus

[9] M. Post. 2018. *A Call for Clarity in Reporting BLEU Scores*. *Proc. WMT*.

[10] M. McCloskey and N. Cohen. 1989. *Catastrophic Interference in Connectionist Networks*. *Psychology of Learning and Motivation*, vol. 24.

[11] A. Chronopoulou et al. 2023. *Language-Family Adapters for Low-Resource Multilingual Translation*. *Proc. ACL*.